Section 8.1 contains an introduction using technologies from Google Cloud Platform (GCP). This part is based on a publicly available two-part workshop by Thoughtworks as well as a data mesh built by Delivery Hero. This platform puts the ownership of the underlying infrastructure into the data-producing teams and utilizes the underlying cloud providers' tooling and integration.

Section 8.2 focuses on Amazon Web Services (AWS) and its object storage, Amazon Simple Storage Service (S3). This part is inspired by a few of the very first data meshes that appeared, like the ones from BMW Group and Zalando. This platform puts the ownership of the infrastructure into the data-producing teams but loosens itself a bit from the underlying cloud provider to allow more freedom.

Section 8.3 explains how Databricks can be used to build up a data mesh. This section is inspired by companies like DPG Media Group, which appeared early on the data mesh stage, but also many more. This section turns the focus away from specific cloud providers, as this technology can be combined with all of the large cloud providers today. This architecture does not put the ownership of the infrastructure into the teams' hands.

Section 8.4 focuses on Apache Kafka, a technology that offers great potential for building data meshes. This example is inspired by a few smaller companies such as Gloo.us that have implemented a data mesh. This last architecture will also not put the ownership of the infrastructure into the teams' hands but will put the data emission and storage of the central system into their hands.

Even though the self-serve data platform is "just" a piece of technology, it has components that enable other elements of the data mesh to work, especially when it comes to (federated computational) governance, data as a product, and (data) domain ownership. So in each section, we will first sketch the architecture, discuss how it works with one or two examples, and then show how these various elements play out in this architecture.

One final note about the word *centralization*, which we will use a lot in this chapter.